

Visweshwaran Ramesh

Postdoctoral Researcher & Data Assimilation - Lead

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Research Profile

I develop data-driven methods for environmental and water systems. I integrate Earth observations and ground observations with hydrological and hydrodynamic modelling using data assimilation and GIS tools. I also work on machine learning and hybrid modelling to improve forecasting and to reduce uncertainty in system prediction.

Work Experience

Postdoctoral Researcher & Data Assimilation - Lead <i>RWTH Aachen University, Data Driven Computing in Civil Engineering (DCC)</i>	2026 – Present <i>Aachen, Germany</i>
Lead Technical Consultant, Data Assimilation & Hydro-Hazard (Part-time) <i>Bhumi Zero-Carbon</i>	2026 – Present <i>Remote</i>
Postdoctoral Researcher & NCEO Fellow <i>University of Reading, Department of Meteorology</i>	2023 – 2025 <i>Reading, UK</i>

Education

Ph.D. in Remote Sensing <i>IIT Bombay</i>	2018 – 2023 <i>Mumbai, India</i>
• Thesis: Towards Improving the Performance of Hydrological Models Using Efficient Assimilation Strategies. • Supervisors: Prof. RAAJ Ramsankaran and Prof. T. I. Eldho. Grade: CGPA 8.91/10.	
M.Tech. in Remote Sensing and GIS <i>NIT Surathkal</i>	2016 – 2018 <i>Mangalore, India</i>
• Thesis: Age-wise Crop Water Requirement of Paddy Fields Under Different Climatic Conditions. • Supervisor: Prof. K. Varija. Grade: CGPA 8.97/10.	
B.E. in Civil Engineering <i>Anna University</i>	2012 – 2016 <i>Chennai, India</i>
• Thesis: Construction and Planning of a Sewage Pipe System for the Chennai Metro Area. • Supervisor: Prof. Sivakumar. Grade: CGPA 8.12/10.	

Projects

1. FLAIR — Flood Forecasting using AI for Regional Sustainability <i>Mentor: Prof. Antara Dasgupta</i> <i>Collaborators: CWRDM, IIT Delhi, Suhora, Floodwaive</i>	2026 – Present <i>IGSTC project</i> <i>RWTH Aachen University</i>
• Developing a deep data assimilation framework that integrates multi-source Earth observations into an AI-based model to forecast regional flood flows and inundation depth in real time. • Building observation operators that link simulated flood states to EO-derived flood extent and water surface elevation within a 2-D convolutional LSTM. • Evaluating forecast skill across two contrasting human-altered catchments, one in India and one in Germany.	
2. Joint State-Parameter Assimilation for the JULES Land Surface Model <i>Mentors: Prof. Sarah L. Dance and Dr. Elizabeth Cooper</i> <i>Collaborators: UKCEH</i>	2023 – 2025 <i>NERC project</i> <i>University of Reading & NCEO</i>
• Built the first joint state-parameter assimilation system for JULES, updating soil moisture state and uncertain soil hydraulic parameters together.	

- Applied a hybrid 4D-EnVar scheme with UKCEH COSMOS-UK field-scale observations, run on the JASMIN HPC facility.
- Tested cyclic assimilation of initial soil moisture and quantified how long the analysis increment persists.

3. Flood Dynamics in East African Wetlands

2025

Mentee: Jake Perrett

HydroJULES Programme

Collaborators: University of Reading, NCEO

UKCEH

- Evaluated CaMa-Flood against MODIS-derived flood extent over the Sudd, South Sudan ($\sim 85,000 \text{ km}^2$), using Kling-Gupta and a bias-insensitive spatial efficiency metric.
- Ran a sensitivity analysis showing inundation is largely insensitive to channel depth and width, but strongly controlled by the channel bifurcation parameters.

4. INCCC — LULC and Climate Change Impacts, Tadri to Kanyakumari

2018 – 2023

Mentors: Prof. RAAJ Ramsankaran and Prof. T. I. Eldho

MoWR project

Collaborators: NITK, CWRDM

IIT Bombay

- Assessed anthropogenic and climate change impacts on the hydrology of the Bharathapuzha basin up to 2100. Used SWAT driven by five downscaled GCMs under RCP 4.5 and RCP 8.5.
- Designed and built the project database and the interactive web interface to be hosted by the Ministry of Water Resources, Government of India.

5. HEC-HMS Runoff Estimation in the Krishna Basin

2017

Mentor: Pankaj Ramji Dhote

M.Tech. thesis internship

IIRS – ISRO

- Estimated runoff over the Krishna basin with HEC-HMS driven by remotely sensed inputs, validated against observed discharge with over 80% correlation.
- Conducted preliminary flood analysis in HEC-RAS to simulate flood scenarios and delineate potential inundation areas.

Awards and Recognition

- **ISH Best PhD Thesis Award**, Indian Society for Hydraulics — certificate and cash prize for outstanding PhD research. 2024
- Invited presentation of NCEO research at the **Royal Society**, London, under the “*Land and Water in a Changing Climate*” theme. 2025
- **Best Presentation Award**, CTCM 2018, for “Age-wise crop water requirement of paddy fields: A case study in Pattambi”. 2018
- **Ph.D. Assistantship**, Ministry of Human Resource Development, Government of India. 2018 – 2023
- **M.Tech Assistantship**, awarded through the GATE examination, Government of India. 2016 – 2018

Mentoring/Supervision

PhD	Ms Greeshma Nair (IIT Bombay) Multivariate data assimilation for an improved distributed GR4J hydro-logical model.	Ongoing
Visiting PhD & M.Sc.	Mr Rajeev Ranjan (IIT Delhi) & Ms Luisa Fuest (RWTH) Particle filter-based flood depth assimilation for improved CaMa-Flood simulations.	Ongoing
M.Tech	Ms Dharani (IIT Madras) Bias-corrected SWOT data assimilation for CaMa-Flood simulations.	Ongoing
B.Sc.	Ms Kim Theisen Trinh (RWTH) Physics-informed emulator of CaMa-Flood for accurate spatiotemporal flood forecasting.	Ongoing
Internship	Mr Jake Perrett (UKCEH) Understanding flood dynamics in East African wetlands.	2025

Teaching Experience

1. Hydroinformatics

RWTH Aachen University, 2026

- Designed and implemented the course evaluation strategy.
- Managed the course structure and examinations, including grading and assessment.

2. RS and GIS for Water Resources Management

IIT Bombay, 2019

- Developed lecture material and delivered lectures on remote sensing applications for hydrological data assimilation.
- Set and evaluated the examination questions for the corresponding Data Assimilation module.

3. Mechanics of Fluids

NIT Surathkal, 2017

- Conducted all laboratory experiments across the full semester.
- Designed, conducted, and evaluated the course examinations.

Skills

Models	JULES, CaMa-Flood, VIC, SWAT, GR4J, HEC suite (HEC-HMS, HEC-RAS)
Data Assimilation	EnKF, particle filter, nudging, hybrid En-Var, 4D-Var
Machine Learning	Deep data assimilation, deep-learning emulators
Programming	Python, MATLAB, C/C++, Bash, Jinja2, HTML/CSS, LaTeX
Software	ArcGIS, QGIS, SNAP, ERDAS, TerrSet, AutoCAD, Affinity Designer
Computing	HPC (JASMIN), Linux cluster management
Systems	Windows, Linux

Invited Talks

1. NCEO/DARC **Data Assimilation Training Course 2025**, under the “Data Assimilation Applications” theme. [\[link\]](#)
2. **5th Annual Land Data Assimilation Community Workshop**, AIMES Land DA Working Group. [\[recording\]](#)

Publications

Journal Articles

1. **Visweshwaran, R.**, Cooper, E., & Dance, S. L. (2026). “Improving land surface model soil moisture estimates through joint state–parameter assimilation of in-situ soil moisture observations”. *Meteorological Applications* [Accepted]. [\[link\]](#)
2. **Visweshwaran, R.**, Patil, A., & Ramsankaran, R. A. A. J. (2025). “Dual correction of rainfall and root zone soil moisture estimates for improving streamflow simulations”. *ARC Geophysical Research*, 1(1). [\[link\]](#)
3. Subbarayan, S., Youssef, Y. M., Singh, L., ..., **Visweshwaran, R.**, & Saqr, A. M. (2025). “SWAT-based prediction of runoff under scenarios of land use/land cover and climate change across Indian agro-climatic zones: Implications for the Sustainable Development Goals”. *Water*, 17(3), Article 458. [\[link\]](#)
4. **Visweshwaran, R.**, Ramsankaran, R. A. A. J., & Eldho, T. I. (2024). “Improving modelled streamflow using time-varying multivariate assimilation of remotely sensed soil moisture and in situ streamflow observations”. *Advances in Water Resources*, 186, Article 104676. [\[link\]](#)
5. **Visweshwaran, R.**, Ramsankaran, R. A. A. J., Eldho, T. I., & Lakshmivarahan, S. (2022). “Sensitivity-based soil moisture assimilation for improved streamflow forecast using a novel forward sensitivity method (FSM) approach”. *Water Resources Research*, 58(1), Article e2021WR031092. [\[link\]](#)
6. **Visweshwaran, R.**, Ramsankaran, R. A. A. J., Eldho, T. I., & Jha, M. K. (2022). “Hydrological impact assessment of future climate change on a complex river basin of the Western Ghats, India”. *Water*, 14(21), Article 3571. [\[link\]](#)

Book Chapters / Manuscripts in Preparation

1. **Visweshwaran, R.**, Ramsankaran, R. A. A. J., & Eldho, T. I. (n.d.). “How do model characteristics impact soil moisture assimilation efficiency? A detailed study using two conceptual rainfall-runoff models” [Manuscript in preparation].

2. Schriewer, E.-L., Kumar, R., Tan, B. T., Fuest, L., Rosas, S. R., Hosch, P., **Visweshwaran, R.**, & Dasgupta, A. (n.d.). "Mapping, measuring, and modelling flood impacts with Earth observation" [Book chapter in preparation].
3. **Visweshwaran, R.**, Ramsankaran, R. A. A. J., Lakshmivarahan, S., & Lewis, J. M. (n.d.). "Dynamic data assimilation in hydrology: A primer" [Book in preparation].

International Conferences and Meetings

1. **Visweshwaran, R.**, & Dasgupta, A. (2026, June 29–July 1). "Deep data assimilation for operational AI-based flood forecasting using multi-source Earth observations". *MLDADS 2026*, DESY Research Centre, Hamburg, Germany. [\[link\]](#)
2. **Visweshwaran, R.**, & Dasgupta, A. (2026, May 3–8). "Towards operational AI-based flood forecasting using deep data assimilation of multi-source Earth observations". *EGU General Assembly 2026*, Vienna, Austria. EGU26-6311. [\[link\]](#)
3. **Visweshwaran, R.**, Cooper, E., & Dance, S. (2025). "Enhancing hydrological forecasting in JULES through EnVar-based soil moisture assimilation of COSMOS observations". *UKCEH Land and Water in a Changing Climate Meeting*, The Royal Society, London, UK.
4. **Visweshwaran, R.**, Cooper, E., & Dance, S. (2025, June 11–13). "Optimising soil moisture assimilation strategies for improved hydrological predictions in the JULES land surface model". *6th IESWG Meeting (IESWG 2025)*, Frankfurt, Germany. [\[link\]](#)
5. **Visweshwaran, R.**, Cooper, E., & Dance, S. (2024, May 20–24). "Improving soil moisture estimates from the JULES land surface model through hybrid 4D-EnVar assimilation of COSMOS-UK soil moisture observations". *12th ADJOINT Workshop*, Lake George, NY, USA. [\[link\]](#)
6. **Visweshwaran, R.**, Dasgupta, A., Ramsankaran, R. A. A. J., & Eldho, T. I. (2023, September 13–15). "Improving the performance of hydrological model forecasts using time-varying multivariate EnKF assimilation". *HEPEX 2023*, SMHI, Norrköping, Sweden. [\[link\]](#)
7. **Visweshwaran, R.**, Ramsankaran, R. A. A. J., & Eldho, T. I. (2023, April 23–28). "Improving streamflow estimates using efficient time-varying multivariate assimilation of soil moisture and streamflow observations". *EGU General Assembly 2023*, Vienna, Austria. EGU23-937. [\[link\]](#)
8. **Visweshwaran, R.**, Ramsankaran, R. A. A. J., & Eldho, T. I. (2022, December 27). "Sequential EnKF assimilation of sensitive soil moisture observations to improve streamflow estimation". *AGU Fall Meeting 2022*. Abstract H24K-06.
9. **Visweshwaran, R.**, Ramsankaran, R. A. A. J., & Eldho, T. I. (2022, December 22–24). "Impact of land use/land cover change on streamflow forecasts using the SWAT model: A case study in a heterogeneous river basin in the Western Ghats". *HYDRO 2022*, Indian Society for Hydraulics, Punjab Engineering College, Chandigarh, India.
10. **Visweshwaran, R.**, Ramsankaran, R. A. A. J., Eldho, T. I., & Lakshmivarahan, S. (2021, December 26–30). "Dynamic data assimilation for improved streamflow forecasts using sensitive soil moisture observations". *AGU Fall Meeting 2021*. Abstract H35W-1296.
11. **Visweshwaran, R.**, & Varija, K. (2018, December 19–21). "Age-wise crop water requirement of paddy fields under different climatic conditions". *HYDRO 2018*, Indian Society for Hydraulics, National Institute of Technology (NIT) Patna, Patna, India.
12. Vijayan, A., Nagaraj, M. K., & **Visweshwaran, R.** (2018, December 19–21,). "Estimation of design flood of a canal regulator at Moolathara using the HEC-HMS model". *HYDRO 2018*, Indian Society for Hydraulics, National Institute of Technology (NIT) Patna, Patna, India.
13. **Visweshwaran, R.**, & Varija, K. (2018). "Age-wise crop water requirement of paddy fields: A case study in Pattambi". *CTCM 2018*, AIJR Publishers.

Poster Presentations

1. **Visweshwaran, R.**, Ramsankaran, R. A. A. J., & Eldho, T. I. (2023, March 15–17). "Improving the performance of hydrological models using efficient assimilation strategies". *IEEE GRSS International Soil Moisture School (ISMS 2023)*, IIT Bombay, Mumbai, India.
2. Cooper, E., **Visweshwaran, R.**, & Dance, S. (2025, September 24). "Land surface data assimilation

community of practice (LSDA-CoP)". *UKCEH Land and Water in a Changing Climate Meeting*, The Royal Society, London, UK.

3. Perrett, J., Folwell, S., Cooper, E., **Visweshwaran, R.**, & Dance, S. (2025, September 24). "Understanding flood dynamics in East African wetlands: Insights from model and observation comparison". *UKCEH Land and Water in a Changing Climate Meeting*, The Royal Society, London, UK.

Blog Posts

1. **Visweshwaran, R.** (2024). "Improving soil moisture prediction using joint state–parameter assimilation in JULES". *Met-DARC Blog*, University of Reading. [\[link\]](#)

Community Building

1. **HDAS-IITB (Hydro Data Assimilation System)**: A research software under development, enabling EnKF-based soil moisture assimilation in the distributed GR4J hydrological model.
2. **Land Surface Data Assimilation Community of Practice (LSDA-CoP)**: Ran the 2023–2025 workshop series bringing together NCEO, UKCEH, the Met Office, ECMWF, NOAA, and the Universities of Reading and Southampton. [\[link\]](#)
3. **"Microwaves from the Sky — the Earth in Radar"**: Delivered an outreach workshop for 20 high-school students at RWTH Aachen. [\[link\]](#)
4. **Journal Peer Review**: Reviewer for ISH Journal of Hydraulic Engineering (Taylor & Francis), Water (MDPI), and Advances in Water Resources (Elsevier).

Field Campaigns

Himachal Pradesh	UAV and DGPS survey at Dhundi station and Sissu.	2023
Hatnur	Sonar-based bathymetric survey of the Krishna River.	2022
Western Ghats	Field surveys across the Bharathapuzha, Chaliyar, and Netravathi river basins.	2019 – 2021

Languages

Tamil (Native) • **English** (Full proficiency) • **Malayalam, Hindi** (Elementary)

References

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